



Network Security and Privacy

NETWORK SECURITY AND PRIVACY

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Network Configuration and Security Implementation

Introduction

This report documents the completion of Assignment 1 for the course Network Security and Privacy (CS 501). The objective of this assignment was to design, configure, and secure a network topology using Cisco Packet Tracer, demonstrating comprehensive understanding of network security principles and practical implementation skills.

The assignment was divided into three main parts:

- **Part A: Network Setup:** Establishing network addressing schemes for Company A, Company B, and DMZ, configuring two websites (public and internal), implementing routing between companies, and setting up DHCP services.
- **Part B: Network Verification:** Proving connectivity between Company B and Company A, and demonstrating access to both websites.

- **Part C: Firewall Rules:** Implementing security policies on the ASA firewall to restrict ICMP traffic while allowing specific web access.

All configurations have been successfully implemented and verified through systematic testing.

Network Topology and IP Addressing Scheme

Network Diagram

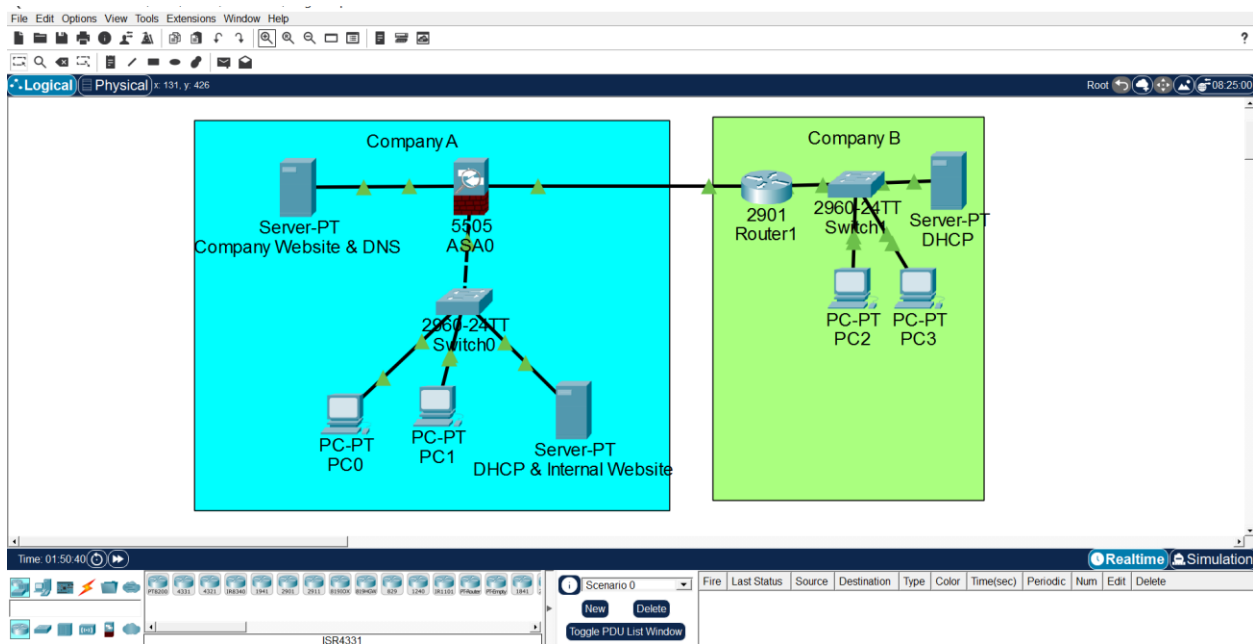


Figure 1: Complete network topology showing Company A, Company B, and DMZ

Network Architecture

The network consists of two separate company networks connected through a DMZ:

- **Company A Network:** Internal LAN with workstations and internal server
- **DMZ (Demilitarized Zone):** Public-facing web server and DNS server
- **Company B Network:** Remote company network with workstations and DHCP server

IP Addressing Scheme

Device	Interface	IP Address	Subnet Mask	Gateway	Purpose
--------	-----------	------------	-------------	---------	---------

Company A					
ASA-CompanyA	VLAN1 (Inside)	192.168.10.1	255.255.255.0	N/A	Company A Gateway
ASA-CompanyA	VLAN2 (Outside)	172.16.1.1	255.255.255.0	N/A	DMZ Gateway
Internal Server	NIC	192.168.10.2	255.255.255.0	192.168.10.1	DHCP + Internal Website
Public Server	NIC	172.16.1.2	255.255.255.0	172.16.1.1	Public Website + DNS
PC0	DHCP	192.168.10.4	255.255.255.0	192.168.10.1	Company A User
PC1	DHCP	192.168.10.3	255.255.255.0	192.168.10.1	Company A User
Company B					
Router-CompanyB	Gi0/0	172.16.1.3	255.255.255.0	172.16.1.1	Connection to ASA
Router-CompanyB	Gi0/1	192.168.20.1	255.255.255.0	N/A	Company B Gateway
DHCP Server	NIC	192.168.20.2	255.255.255.0	192.168.20.1	DHCP Server
PC2	DHCP	192.168.20.4	255.255.255.0	192.168.20.1	Company B User
PC3	DHCP	192.168.20.3	255.255.255.0	192.168.20.1	Company B User

Table 1: Complete IP addressing scheme for all devices

Network Segmentation

The network is segmented into three distinct zones:

1. **Company A Internal Zone (192.168.10.0/24):** Contains internal workstations and the internal web server with confidential information
2. **DMZ Zone (172.16.1.0/24):** Contains the public-facing web server and DNS server, isolated from internal networks
3. **Company B Zone (192.168.20.0/24):** Contains remote company workstations and DHCP server

This segmentation follows security best practices by isolating critical internal resources from publicly accessible servers.

Part A: Network Setup Configuration

Company A ASA5505 Firewall Configuration

The ASA5505 serves as the primary firewall and gateway for Company A, providing security and routing between networks.

VLAN Configuration

Two VLANs were configured on the ASA:

- **VLAN 1 (Inside):** Company A internal network (192.168.10.0/24)
- **VLAN 2 (Outside):** DMZ network (172.16.1.0/24)

```
ASA-CompanyA(config)#configure terminal
ASA-CompanyA(config)#no access-group OUTSIDE_IN in interface outside
ASA-CompanyA(config)#interface Vlan1
ASA-CompanyA(config-if)# nameif inside
ASA-CompanyA(config-if)# ip address 192.168.10.1 255.255.255.0
ASA-CompanyA(config-if)# no shutdown
ASA-CompanyA(config-if)#
ASA-CompanyA(config-if)#interface Vlan2
ASA-CompanyA(config-if)# nameif outside
ASA-CompanyA(config-if)# ip address 172.16.1.1 255.255.255.0
ASA-CompanyA(config-if)# no shutdown
ASA-CompanyA(config-if)#
```

Figure 2: Configuration Commands

Interface Assignment

Physical interfaces were assigned to respective VLANs:

- Ethernet0/0-1: Assigned to VLAN 2 (outside)
- Ethernet0/2-7: Assigned to VLAN 1 (inside)

```
ASA-CompanyA(config-if)#route outside 0.0.0.0 0.0.0.0 172.16.1.3
ASA-CompanyA(config)#route inside 192.168.20.0 255.255.255.0 172.16.1.3
```

Figure 3: ASA interface VLAN assignments

- Default route sends all unknown traffic to Company B router
- Specific route for Company B network (192.168.20.0/24) via 172.16.1.3

Static Routing Configuration

Static routes were configured to enable communication between networks:

```
ASA-CompanyA(config)#route outside 0.0.0.0 0.0.0.0 172.16.1.3
ASA-CompanyA(config)#route inside 192.168.20.0 255.255.255.0 172.16.1.3
ASA-CompanyA(config)#
```

- Default route sends all unknown traffic to Company B router
- Specific route for Company B network (192.168.20.0/24) via 172.16.1.3

DHCP Configuration on ASA

DHCP services were enabled on the inside interface for Company A workstations:

```
ASA-CompanyA(config)#
ASA-CompanyA(config)#configure terminal
ASA-CompanyA(config)#no dhcpd address 192.168.10.100-192.168.10.131 inside
ASA-CompanyA(config)#no dhcpd enable inside
ASA-CompanyA(config)#write memory
Building configuration...
Cryptochecksum: 103943ca 16ec3bb3 7bc75584 4ad1320f

1497 bytes copied in 1.526 secs (980 bytes/sec)
[OK]
ASA-CompanyA(config)#
```

Copy

Paste

Figure 4: ASA DHCP CONFIGURATION

ASA Interface Verification

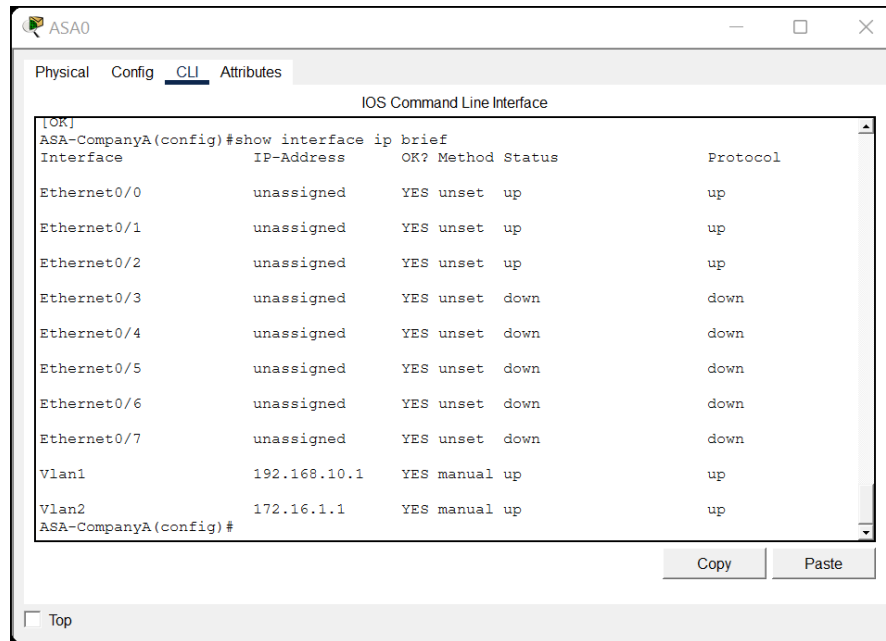


Figure 5: ASA interface IP verification

The output confirms:

- VLAN1 (inside): 192.168.10.1/24
- VLAN2 (outside): 172.16.1.1/24
- Both interfaces are up and operational

Company A Internal Server (DHCP + Internal Website)

The internal server in Company A serves two critical functions:

1. **DHCP Server:** Provides IP addresses to Company A workstations
2. **Internal Web Server:** Hosts the confidential internal company page

IP Configuration

Static IP Configuration:

The screenshot shows the 'DHCP & Internal Website' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is active, showing 'Static' configuration for both IPv4 and IPv6. The IPv4 configuration includes an IP address of 192.168.10.2, a subnet mask of 255.255.255.0, a default gateway of 192.168.10.1, and a DNS server of 172.16.1.2. The IPv6 configuration is also set to 'Static' with a link-local address of FE80::201:42FF:FE0C:1058.

IP Configuration	
☐ DHCP ☒ Static	
IPv4 Address	192.168.10.2
Subnet Mask	255.255.255.0
Default Gateway	192.168.10.1
DNS Server	172.16.1.2
IPv6 Configuration	
☐ Automatic ☒ Static	
IPv6 Address	FE80::201:42FF:FE0C:1058
Link Local Address	
Default Gateway	
DNS Server	

Figure 6: Internal server static IP configuration

DHCP Server Configuration

The DHCP server was configured with the following pool:

The screenshot shows the 'DHCP & Internal Website' configuration window with the 'Services' tab selected. The 'DHCP' service is configured for the 'FastEthernet0' interface. The 'CompanyA_Pool' is defined with a default gateway of 192.168.10.1, a DNS server of 172.16.1.2, and a start IP address of 192.168.10.100. The pool also has a subnet mask of 255.255.255.0, a maximum number of users of 50, and TFTP and WLC addresses of 0.0.0.0.

SERVICES	
☒ HTTP ☐ DHCP	
☐ DHCPv6 ☐ TFTP	
☐ DNS ☐ SYSLOG	
☐ AAA ☐ NTP	
☐ EMAIL ☐ FTP	
☐ IoT ☐ VM Management	
☐ Radius EAP ☐ PRP	

DHCP	
Interface	FastEthernet0
Service	☒ On ☐ Off
Pool Name	CompanyA_Pool
Default Gateway	192.168.10.1
DNS Server	172.16.1.2
Start IP Address :	192 168 10 100
Subnet Mask:	255 255 255 0
Maximum Number of Users :	50
TFTP Server:	0.0.0.0
WLC Address:	0.0.0.0

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
CompanyA_Pool	192.168.10.1	172.16.1.2	192.168.10.100	255.255.255.0	50	0.0.0.0	0.0.0.0

Figure 7: Internal server DHCP pool configuration

Internal Website Configuration

The internal website was configured with a confidential page, clearly marked for authorized personnel only:

HTML Code:

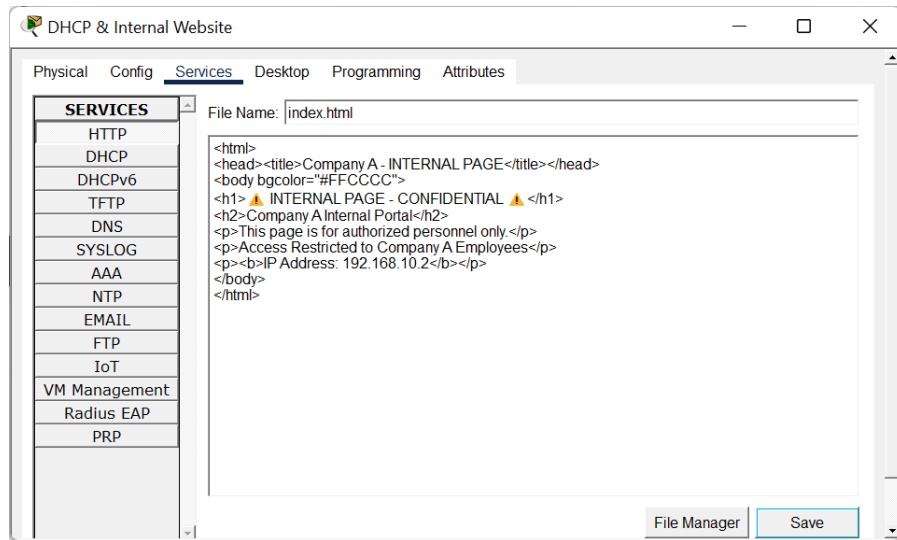


Figure 8: Internal website HTML configuration

Company A Public Website & DNS Server

The public server is placed in the DMZ and serves as both the public-facing web server and DNS server for the network.

IP Configuration

Static IP Configuration:

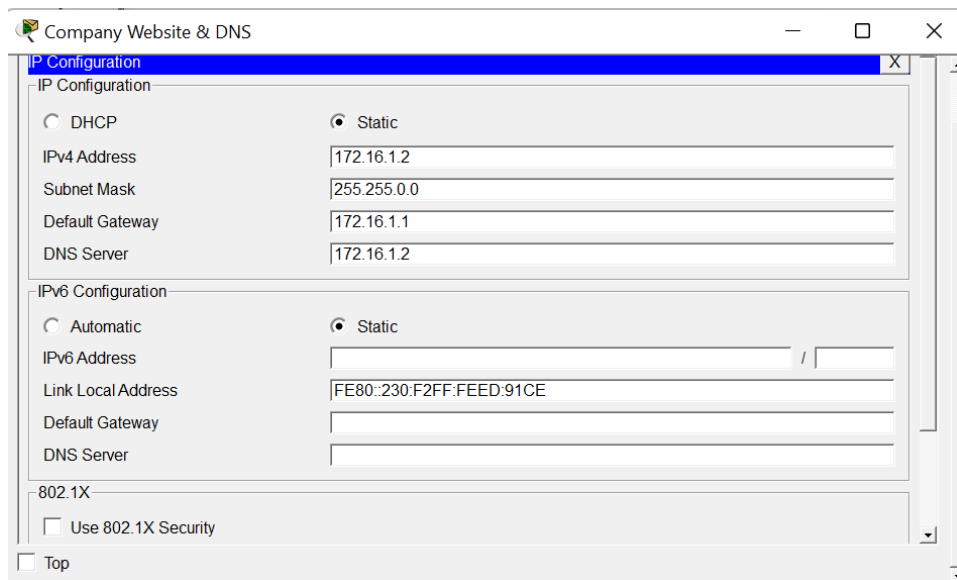


Figure 9: Public server static IP configuration in DMZ

Public Website Configuration

The public website was configured with a welcoming page for external visitors:

HTML Code:

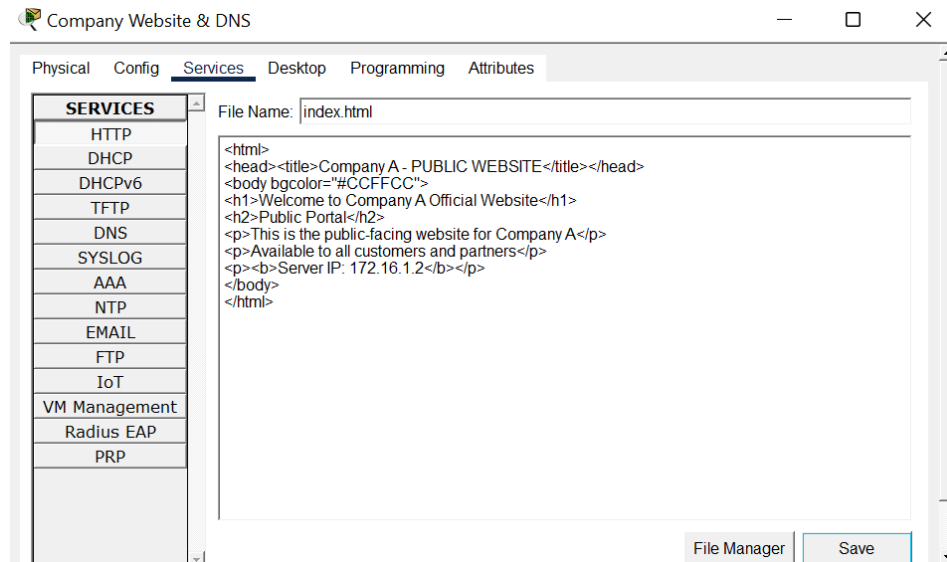


Figure 10: Public website HTML configuration

DNS Server Configuration

DNS services were enabled to resolve domain names to IP addresses:

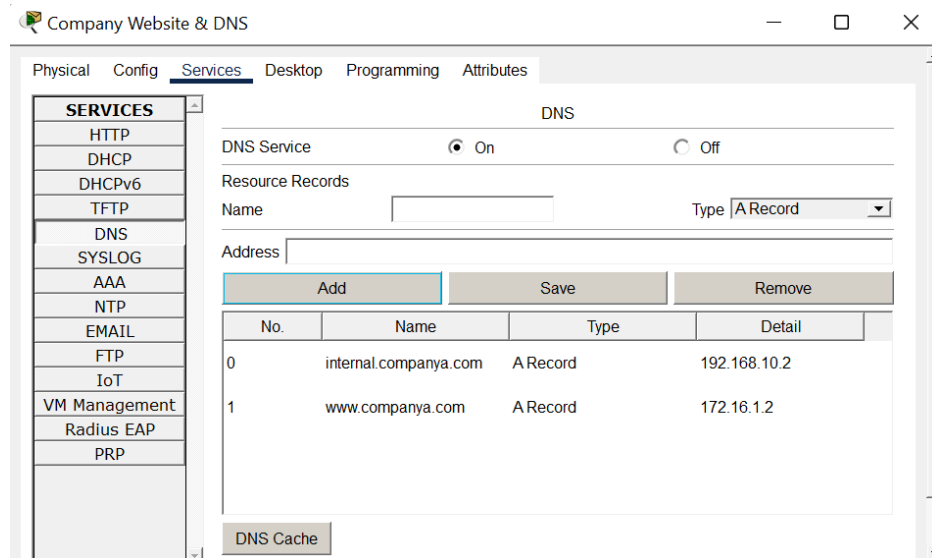


Figure 11: DNS server configuration with A records

Company B Router Configuration

The Company B router (2901) connects the Company B LAN to the ASA firewall and provides gateway services.

Basic Router Configuration

- Interface Configuration
- GigabitEthernet0/0 (Connection to ASA)
- GigabitEthernet0/1 (Connection to Company B LAN)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname Router-CompanyB
Router-CompanyB(config)#interface gigabitEthernet0/0
Router-CompanyB(config-if)#description Connection to ASA
Router-CompanyB(config-if)#ip address 172.16.1.3 255.255.255.0
Router-CompanyB(config-if)#no shutdown

Router-CompanyB(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router-CompanyB(config-if)#exit
Router-CompanyB(config)#interface gigabitEthernet0/1
Router-CompanyB(config-if)#description Connection to Company B LAN
Router-CompanyB(config-if)#ip address 192.168.20.1 255.255.255.0
Router-CompanyB(config-if)#no shutdown
Router-CompanyB(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router-CompanyB(config-if)#exit
Router-CompanyB(config)#ip route 192.168.10.0 255.255.255.0 172.16.1.1
Router-CompanyB(config)#ip route 172.16.1.0 255.255.255.0 172.16.1.1
Router-CompanyB(config)#end
Router-CompanyB#
%SYS-5-CONFIG_I: Configured from console by console

Router-CompanyB#write memory
Building configuration...
[OK]
Router-CompanyB#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

Figure 12: Company B router interface configuration

Static Route Configuration

Static routes were added to reach Company A networks:

```
Router-CompanyB(config)#ip route 192.168.10.0 255.255.255.0 172.16.1.1
```

```
Router-CompanyB(config)#ip route 172.16.1.0 255.255.255.0 172.16.1.1
```

Routing Table Verification

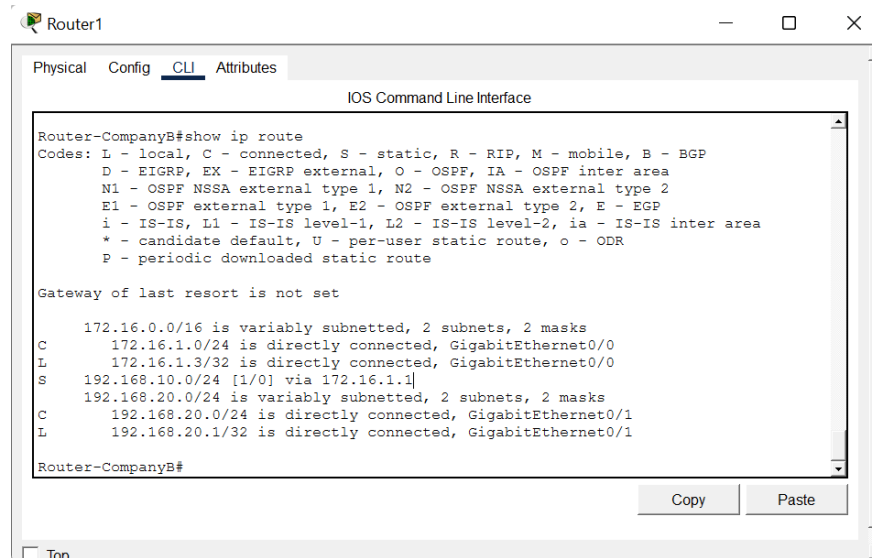


Figure 13: Company B router routing table

The routing table confirms:

- Directly connected networks: 172.16.1.0/24 and 192.168.20.0/24
- Static route to 192.168.10.0/24 via 172.16.1.1

Company B DHCP Server Configuration

The DHCP server in Company B provides IP addresses to workstations in the Company B network.

IP Configuration

The screenshot shows the 'DHCP' configuration window with the 'Desktop' tab selected. The 'IP Configuration' section is expanded, showing 'Static' configuration for both IPv4 and IPv6. The IPv4 settings are: IPv4 Address: 192.168.20.2, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.20.1, and DNS Server: 172.16.1.2. The IPv6 settings are: IPv6 Address: (empty), Link Local Address: FE80::250:FFF:FE00:31BD, Default Gateway: (empty), and DNS Server: (empty). The '802.1X' section is also visible with 'Use 802.1X Security' unchecked.

Figure 14: Company B DHCP server static IP configuration

DHCP Pool Configuration

The DHCP server was configured with the following pool:

The screenshot shows the 'DHCP' configuration window with the 'Services' tab selected. The 'DHCP' section is expanded, showing the 'FastEthernet0' interface with 'Service' set to 'On'. The 'Pool Name' is 'CompanyB_Pool'. The 'Default Gateway' is 192.168.20.1 and the 'DNS Server' is 172.16.1.2. The 'Start IP Address' is 192.168.20.100 and the 'Subnet Mask' is 255.255.255.0. The 'Maximum Number of Users' is 50. The 'TFTP Server' is 0.0.0.0 and the 'WLC Address' is 0.0.0.0. Below the configuration fields are 'Add', 'Save', and 'Remove' buttons. A table at the bottom lists the configured pool.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
CompanyB_Pool	192.168.20.1	172.16.1.2	192.168.20.100	255.255.255.0	50	0.0.0.0	0.0.0.0

Figure 15: Company B DHCP pool configuration

Switch Configurations

Company A Switch (Switch0)

The switch in Company A connects workstations and the internal server to the ASA.

VLAN Configuration:

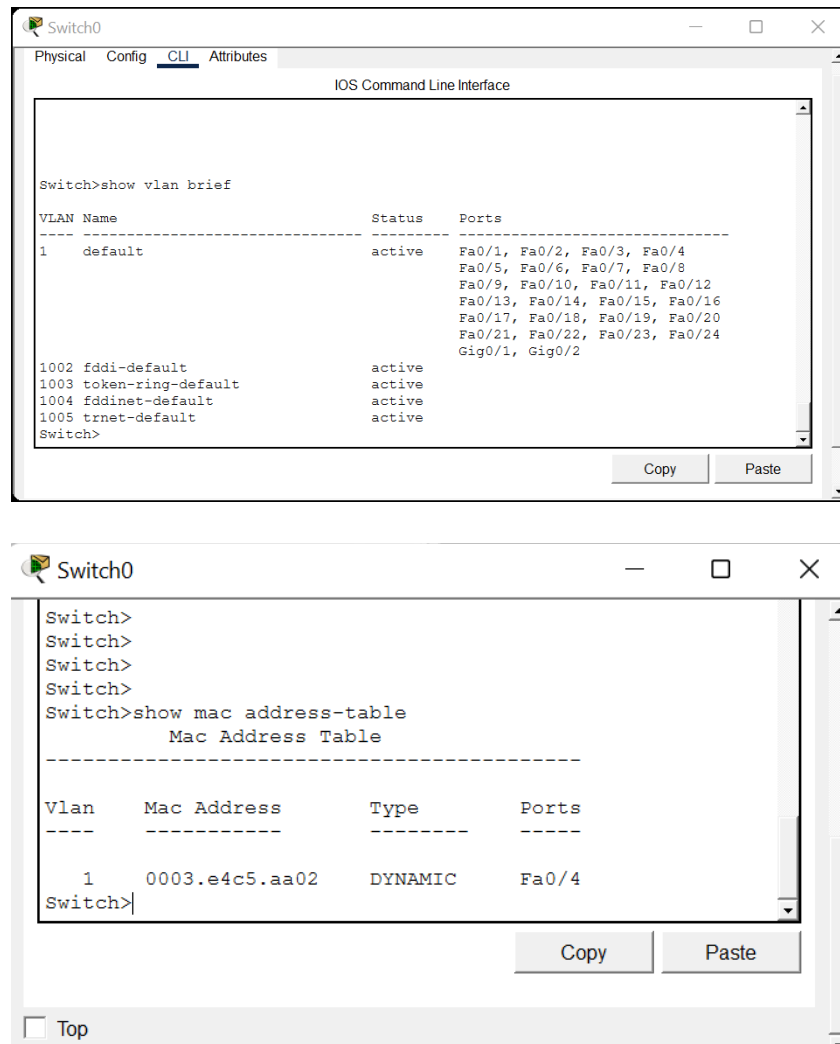


Figure 16: Company A switch VLAN configuration & MAC address table

Company B Switch (Switch1)

The switch in Company B connects workstations and the DHCP server to the router.

VLAN Configuration:

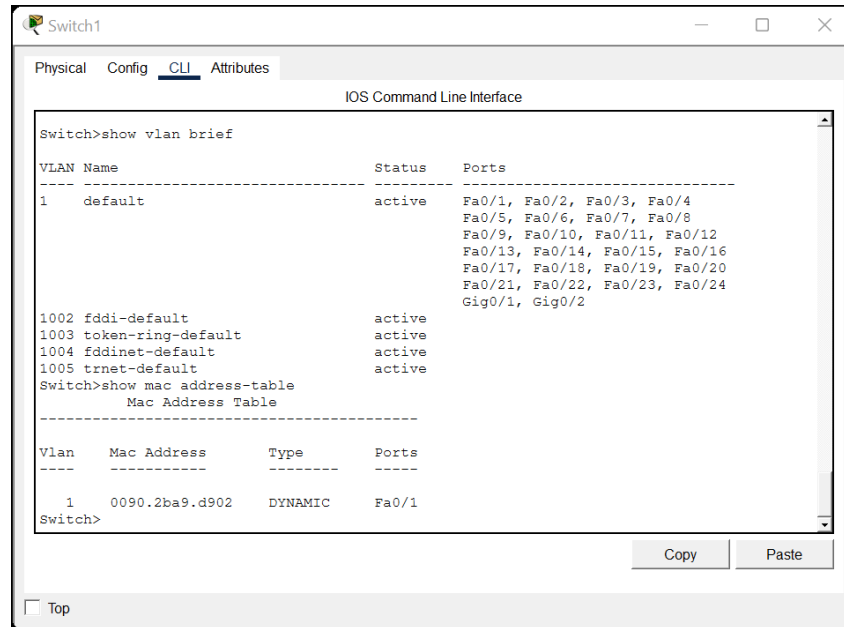


Figure 17: Company B switch VLAN configuration & MAC address table

PC DHCP Configuration

All workstations were configured to obtain IP addresses automatically via DHCP.

Company A Workstations

PC0 IP Configuration:

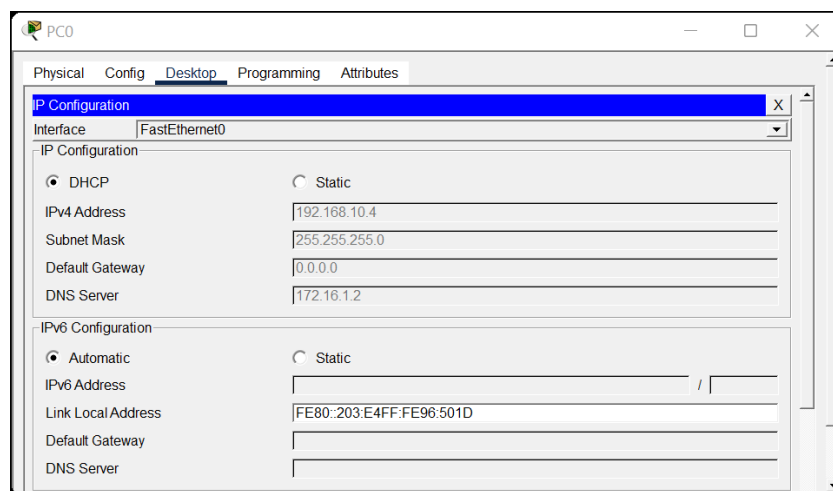


Figure 18: PC0 DHCP-acquired IP configuration

PC1 IP Configuration:

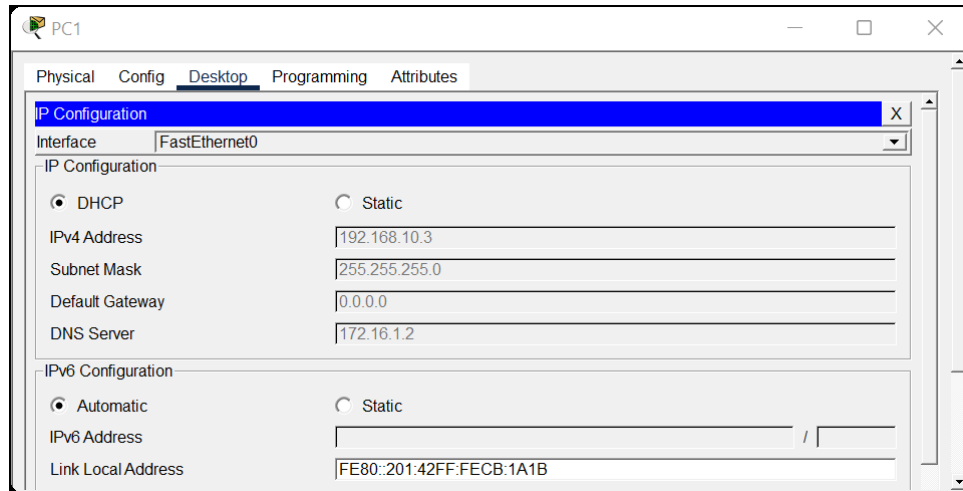


Figure 19: PC1 DHCP-acquired IP configuration

Company B Workstations

PC2 IP Configuration:

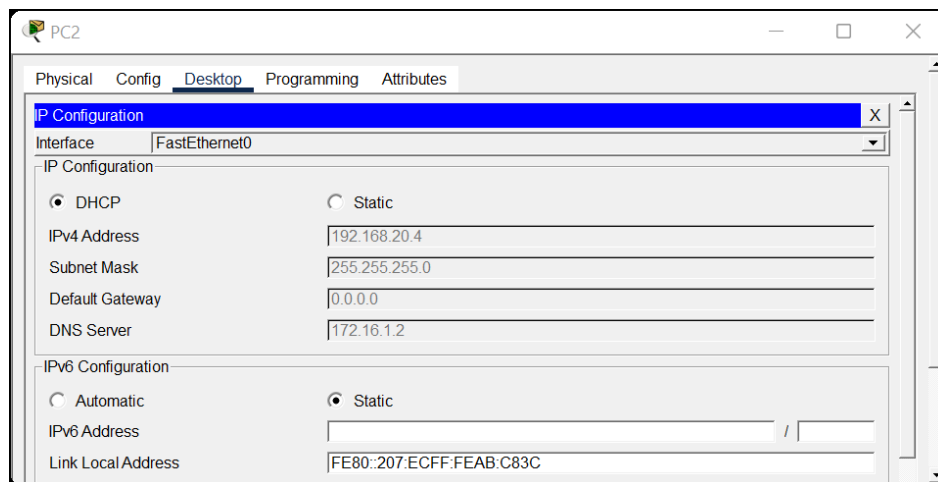


Figure 20: PC2 DHCP-acquired IP configuration

PC3 IP Configuration:

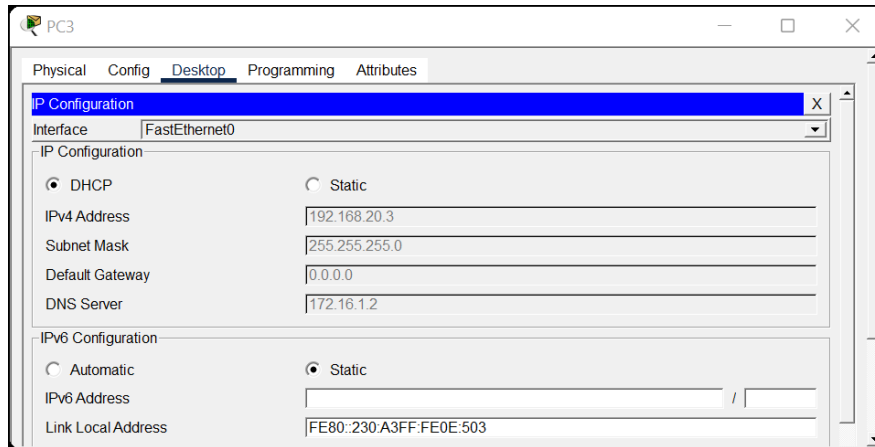


Figure 21: PC3 DHCP-acquired IP configuration

Part B: Network Verification (Before Firewall)

For the purpose of demonstrating full connectivity as required in Part B, the firewall was temporarily disabled using the command `no access-group OUTSIDE_IN` in interface `outside` on the ASA. This allows us to prove that the network infrastructure itself is functioning correctly before security policies are applied.

Connectivity Testing

Ping from Company B Router to ASA

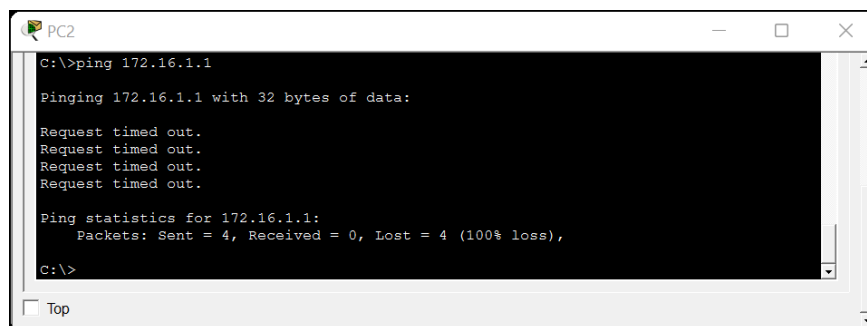
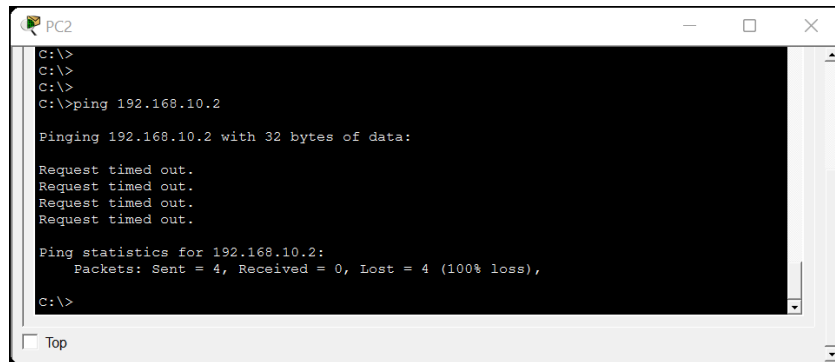


Figure 22: Router-Company B ping to ASA (172.16.1.1) - SUCCESSFUL

The successful ping confirms Layer 3 connectivity between Company B router and the ASA firewall.

Ping from Company B PC to Company A Internal Server



```
C:\>
C:\>
C:\>
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

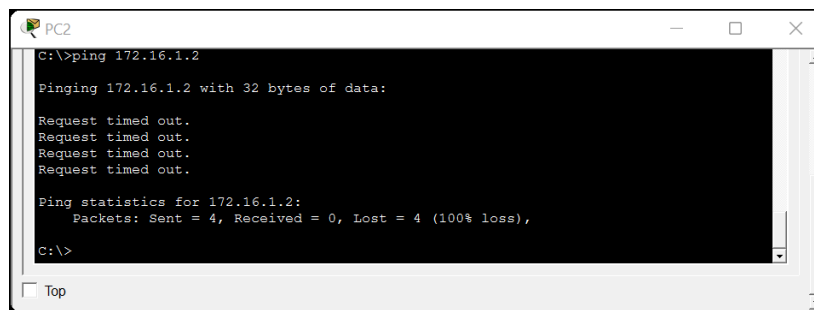
Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 23: PC2 ping to Company A internal server (192.168.10.2) - SUCCESSFUL

This demonstrates end-to-end connectivity from Company B workstation to Company A internal server.

Ping from Company B PC to Company A Public Server



```
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 24: PC2 ping to Company A public server (172.16.1.2) - SUCCESSFUL

This confirms connectivity from Company B to the DMZ.

Website Accessibility Testing

Accessing Public Website from Company B

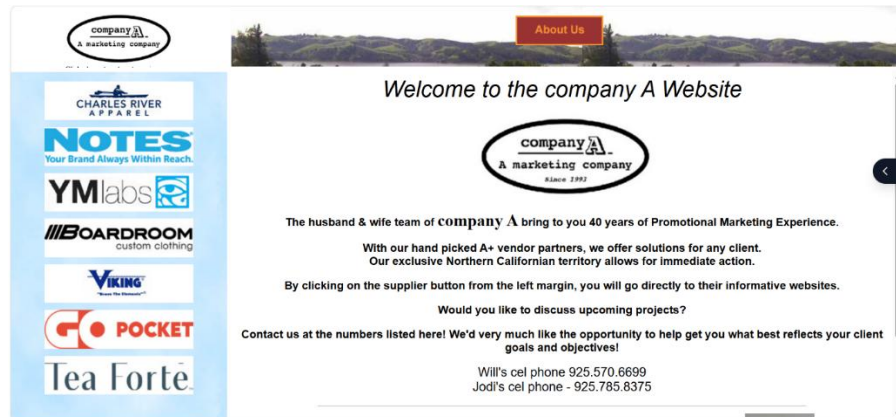


Figure 25: PC2 accessing public website (<http://172.16.1.2>) - SUCCESSFUL

The public website loads successfully, proving that Company B can reach the DMZ web server.

Accessing Internal Website from Company B

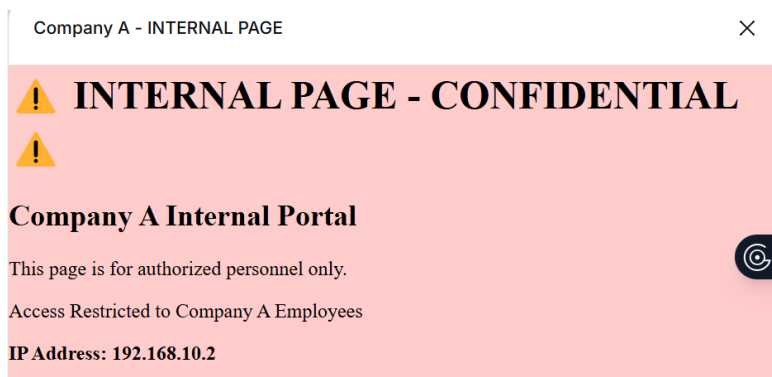


Figure 26: PC2 accessing internal website (<http://192.168.10.2>) - SUCCESSFUL

The internal website loads successfully, proving that without firewall restrictions, Company B can access internal Company A resources.

Part C: ASA Firewall Rules Configuration

Firewall Access Control Lists (ACL)

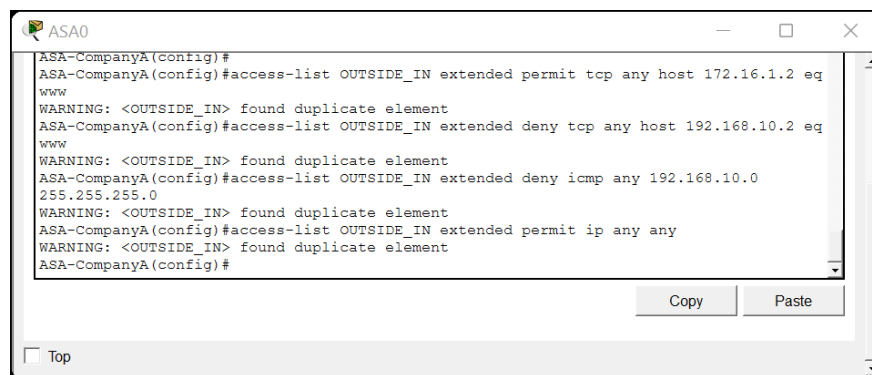
The ASA firewall was configured with an access list named OUTSIDE_IN to enforce security policies as required by the assignment.

ACL Requirements

The assignment specified three specific requirements:

1. **Company B cannot ping Company A hosts (Deny ICMP)**
2. **Company B can access the company website (Permit HTTP to public server)**
3. **Company B cannot access the Internal page (Deny HTTP to internal server)**

ACL Configuration



```
ASA-CompanyA(config)#
ASA-CompanyA(config)#access-list OUTSIDE_IN extended permit tcp any host 172.16.1.2 eq
www
WARNING: <OUTSIDE_IN> found duplicate element
ASA-CompanyA(config)#access-list OUTSIDE_IN extended deny tcp any host 192.168.10.2 eq
www
WARNING: <OUTSIDE_IN> found duplicate element
ASA-CompanyA(config)#access-list OUTSIDE_IN extended deny icmp any 192.168.10.0
255.255.255.0
WARNING: <OUTSIDE_IN> found duplicate element
ASA-CompanyA(config)#access-list OUTSIDE_IN extended permit ip any any
WARNING: <OUTSIDE_IN> found duplicate element
ASA-CompanyA(config)#
```

Figure 27: access-list Configurations

ACL Application

The access list was applied to the outside interface in the inbound direction:

```
ASA-CompanyA(config)#access-group OUTSIDE_IN in interface outside
ASA-CompanyA(config)#
```

ACL Logic Explanation

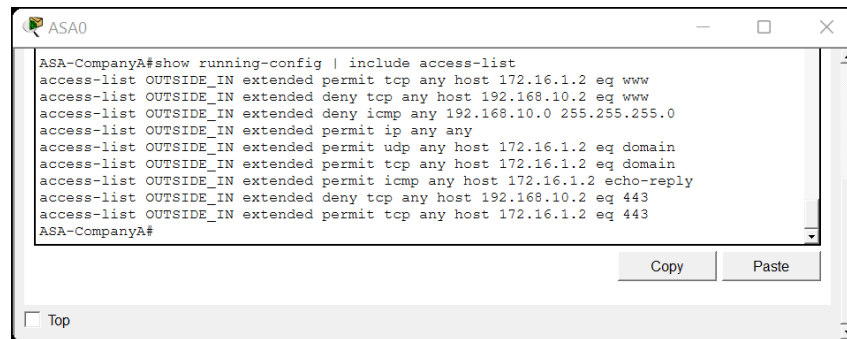
Line	Rule	Action	Purpose
------	------	--------	---------

1	permit tcp any host 172.16.1.2 eq www	ALLOW	Company B can access public website
2	deny tcp any host 192.168.10.2 eq www	DENY	Company B cannot access internal page
3	deny icmp any 192.168.10.0 255.255.255.0	DENY	Company B cannot ping Company A hosts
4	permit ip any any	ALLOW	Allow all other traffic (implicit deny otherwise)

Table 2: ASA firewall ACL configuration

Verification of Firewall Rules

ACL Verification



```

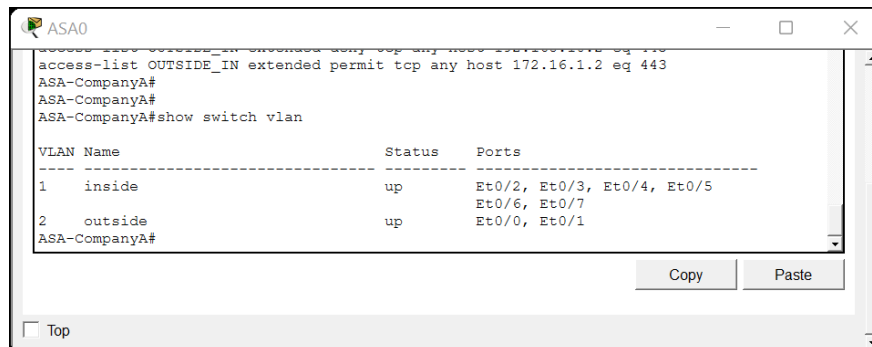
ASA-CompanyA#show running-config | include access-list
access-list OUTSIDE_IN extended permit tcp any host 172.16.1.2 eq www
access-list OUTSIDE_IN extended deny tcp any host 192.168.10.2 eq www
access-list OUTSIDE_IN extended deny icmp any 192.168.10.0 255.255.255.0
access-list OUTSIDE_IN extended permit ip any any
access-list OUTSIDE_IN extended permit udp any host 172.16.1.2 eq domain
access-list OUTSIDE_IN extended permit tcp any host 172.16.1.2 eq domain
access-list OUTSIDE_IN extended permit icmp any host 172.16.1.2 echo-reply
access-list OUTSIDE_IN extended deny tcp any host 192.168.10.2 eq 443
access-list OUTSIDE_IN extended permit tcp any host 172.16.1.2 eq 443
ASA-CompanyA#

```

Figure 28: show running-config | include access-list

The hit counts in the ACL output will show how many packets have matched each rule, confirming that the rules are active and being used.

ASA VLAN Status



```

ASA-CompanyA#show switch vlan
VLAN Name                Status    Ports
-----
1    inside                 up        Et0/2, Et0/3, Et0/4, Et0/5
2    outside                 up        Et0/6, Et0/7
ASA-CompanyA#

```

Figure 29: ASA VLAN status verification

The output confirms:

- VLAN 1 (inside): up with ports Et0/2-7
- VLAN 2 (outside): up with ports Et0/0-1

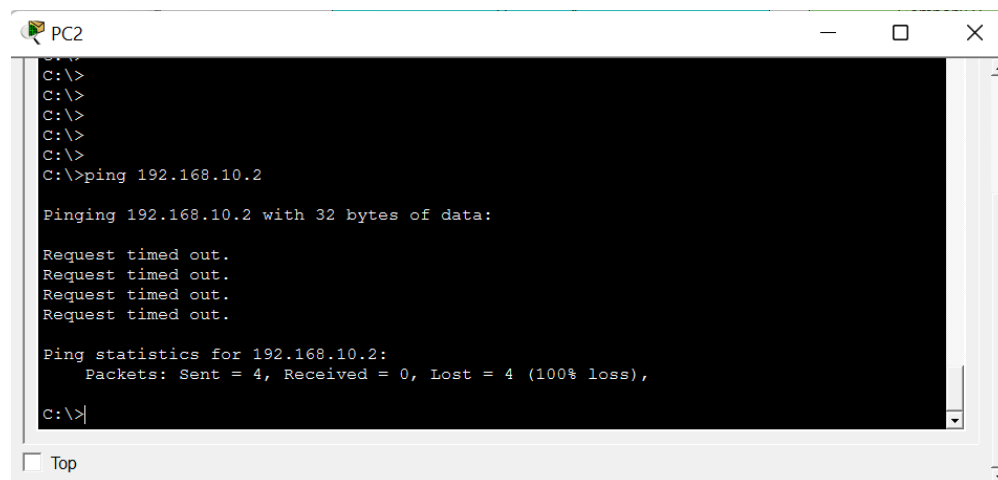
Part B & C Combined Verification (With Firewall)

The following tests were conducted with the firewall fully enabled (access-group OUTSIDE_IN in interface outside). These results demonstrate that the firewall rules are working as intended.

Ping Test from Company B to Company A (Should FAIL)

Test: From PC2 (Company B) ping the Company A internal server (192.168.10.2)

Expected Result: Request timed out (packets dropped by firewall)



```
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>|
```

Figure 30: Ping from PC2 to 192.168.10.2 - REQUEST TIMED OUT ✓

Analysis: The ping fails as expected because the firewall rule deny icmp any 192.168.10.0 255.255.255.0 blocks all ICMP traffic to Company A network. This confirms Requirement 1 is satisfied.

Public Website Access Test (Should WORK)

Test: From PC2 (Company B) access the public website at <http://172.16.1.2>

Expected Result: Public website loads successfully

PC2 accessing public website (<http://172.16.1.2>) - SUCCESSFUL ✓

Analysis: The public website loads successfully because the firewall rule permit tcp any host 172.16.1.2 eq www allows HTTP traffic to the public server. This confirms Requirement 2 is satisfied.

Internal Website Access Test (Should FAIL)

Test: From PC2 (Company B) access the internal website at <http://192.168.10.2>

Expected Result: Request timeout or browser unable to connect

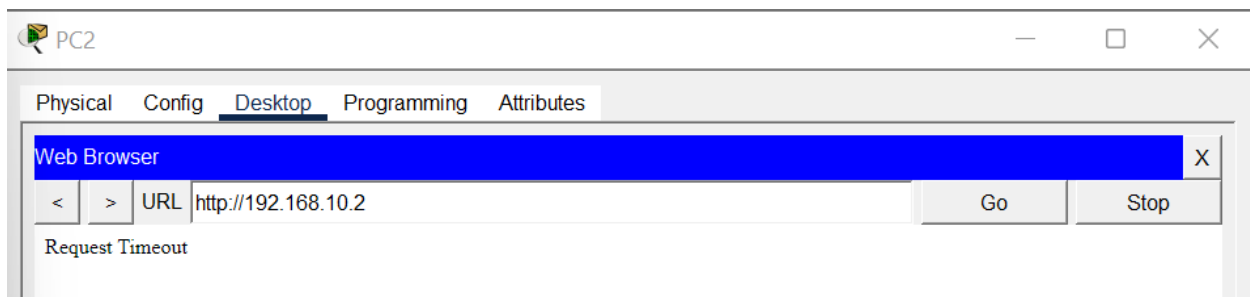


Figure 31: PC2 accessing internal website (<http://192.168.10.2>) - REQUEST TIMEOUT ✓

Analysis: The internal website fails to load because the firewall rule denies tcp any host 192.168.10.2 eq www blocks HTTP traffic to the internal server. This confirms Requirement 3 is satisfied.

Conclusion

This assignment successfully demonstrated the design, configuration, and security implementation of a multi-zone network topology using Cisco Packet Tracer. The network functions exactly as specified in the assignment requirements. All devices are properly configured, routing works correctly, DHCP provides IP addresses automatically, and the ASA firewall enforces the required security policies. This implementation demonstrates a solid understanding of network security principles and practical configuration skills.